

# An Outlook on using Packet Sampling in Flow-based C2 TLS Malware Traffic Detection

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**Abstract**—Packet sampling plays an important role in keeping storage and processing requirements at a manageable level in network management. However, because it reduces the amount of available information, it can also reduce the performance of some related tasks, such as detecting security events. In this context, this work explores how packet sampling impacts machine learning-based tasks, in particular, flow-based C2 TLS malware traffic detection using a deep neural network. Based on a proposed lightweight sampling scheme, the ongoing results show a small reduction in classification accuracy compared with analysing all the traffic, while reducing in 10 fold the number of packets processed.

## I. INTRODUCTION

Infected machines typically communicate with a malicious Command and Control (C2) server to receive further attack orders or to upload stolen information. In order to avoid detection by payload inspection, C2 traffic is rapidly migrating to encrypted communications, namely, resorting to TLS given its widespread use. Currently, TLS C2 traffic is commonly detected through server-side certificate blacklisting. Yet, with free TLS certificate providers, attackers can seamlessly use new certificates for their C2 servers and avoid this detection. A strategy aiming to overcome this problem is to use detectors based on traffic and traffic features to complement certificate blacklisting.

Traffic-based detectors rely on packet processing to generate classification features. However, due to the massive amount of data traversing current networks, administrators often rely on packet sampling techniques to mitigate the overhead of packet processing and storage. This strategy consists of performing monitoring tasks based only on a subset of packets reaching a measurement point. When applied to C2 traffic detection, sampling can change traffic feature values which, in turn, may affect the accuracy of traffic feature-based detection processes.

In this context, this work provides ongoing results on the impact of packet sampling in C2 traffic detection. In particular, a deep learning classifier was trained with TCP flow features from sampled and non-sampled traffic for further performance comparison. The ongoing results show a small reduction in classification accuracy when analysing only TCP handshake and finish packets plus the two initial packets in each flow

direction, which reduces the number of processed packets up to an average of 10 times. These results are promising and indicate that exploring the flexibility of different sampling techniques may yield better classifier results while reducing even more the amount of packets required. This work also reports on predicting non-sampled feature values from sampled ones using an autoencoder-like structure, yet these features are still far from successfully detecting C2 traffic this way.

This paper is organized as follows: Sec. II discusses some of the related work in this area; the proposed sampling scheme and the detection approach are presented in Sec. III; the dataset used to measure detection accuracy is described in Sec. IV; the ongoing results are discussed in Sec. V; Sec. VI extends the results by discussing alternative approaches toward combining packet sampling and machine learning; and some final remarks are summarised in Sec. VII.

## II. RELATED WORK

In network monitoring, traffic sampling as a strategy to reduce processing overhead is used at two levels of granularity: (i) *flow-level*, in which the measurement point selects all packets belonging to a specific flow to compose a sample, by applying a filter to all traffic or (ii) *packet-level*, where all packets are considered single independent entities, regardless of which flow they belong to. Selecting packets to a sample might follow a deterministic, probabilistic, or adaptive approach [1].

In the deterministic approach, the process of packet selection is ruled by a deterministic function which imposes a fixed sampling frequency based on the packet position or content. The random approach is ruled by a probabilistic function that can be uniform or non-uniform. Lastly, within an adaptive approach, the selection process is able to change the packet selection criterion during the course of measurements in response to the traffic behavior, expected accuracy or resource constraints [2].

Different packet sampling approaches have been successfully used in specific network monitoring activities. For instance, deterministic schemes have shown to be adequate for traffic accounting [3] while probabilistic and adaptive approaches are suitable for monitoring QoS parameters and SLA compliance [4]. When applied to security-related tasks, such as traffic anomaly detection [5], DDoS attacks [6], and

intrusion [7] the recommendations are more variable. The reason is that each task requires observing a particular portion of the traffic, and different sampling techniques impact what can be measured differently.

In this context, a literature review did not identify related work that assessing the impact of sampling on flow-based encrypted malware C2 traffic detection. Yet there are many malware traffic detection approaches for which different sampling techniques may or may not be suitable. For example, [8] takes standard and enhanced sets of flow features and uses several shallow machine learning techniques to detect malicious flows captured from an enterprise network. Another example [9] uses raw packet bytes to detect different types of malicious traffic (although not encrypted malware C2) from publicly available sources, feeding the first  $n$  bytes of each of the first  $m$  packets to a neural network which is then trained to detect malicious traffic.

Although [8] and [9] do not consider sampled traffic, the extended set of flow features in [8] differs significantly from this work given that it includes sizes of initial packets and TLS metadata, and [9] includes the contents of the initial packets of each flow which is very different from flow features. In both approaches, packets not selected for a sample may have a significant impact in their detection accuracy. Using image processing as a metaphor, in these cases where the information that is sampled out does not reach the neural network it would lead to black pixels in the image, whereas in this proposal where the feature values change with sampling, it would lead to a blurred image. Therefore, the sampling approach proposed in this paper may not apply directly in these and other malicious traffic detection approaches.

Specifically for detectors based on flow features such as Tstat [10] and the standard set of features in [8], a relevant body of work exists for flow-level sampling. However, most of it focuses on sampling flows and not sampling packets within each flow, often for detecting DoS events and not for classifying individual flows. Specific papers that do focus on sampling packets within flows such as [11] take aim at elephant flows which is likely not the case in C2 traffic that is intended to remain undetected.

### III. DETECTION APPROACH

The process of detecting C2 traffic explored in this work consists of firstly sampling packets from the monitored network. Then, the sampled data is processed by the Tstat analyzer [10] in order to extract features describing each flow within the sample. Lastly, these features are processed by a machine learning classifier trained to detect malicious traffic.

#### A. Sampling technique

The machine learning-based C2 traffic classification approach used in this work requires features representing full flows. Therefore, the sampling technique should include, at least, all packets containing the TCP handshake flags. To achieve this, a two stage sampling scheme is defined and tested

whose aim is to balance the accuracy in C2 detection with the reduction of the amount of data required from the network.

- *Stage 1* consists of a deterministic sampling scheme based on packet content set to select and collect all TCP packets with SYN, FIN and RST flags active, and the two packets that immediately follow these flagged packets in the same direction. These two initial packets contain TLS handshake records and allow Tstat to identify the flow as TLS.
- *Stage 2* samples the remaining packets of each flow using a classical and lightweight deterministic technique that consists of selecting and capturing packets in a systematic pattern, namely *n-out-of-N*, meaning that  $n$  packets are collected for the sample from a sequential stream of  $N$  packets [1]. Stage 2 considers that including more of the flow's packets in the sample beyond handshake packets is more representative of relevant features that may be required by the classifier.

In the rest of the paper, the term 'Full' represents the non-sampled traffic, 'Det.-1/m' represents sampled flows with Stage 2 values  $n = 1$  and  $N = m$  which indicates that  $1 - out - of - m$  packets are selected for the sample, and 'Det.-0' indicates that Stage 2 is not applied and thus only the packets defined in Stage 1 are selected for further classification. If not specifically mentioned, results are for the Stage 1-only 'Det.-0' approach.

#### B. Classifier

The classifier is an 86 input, 3-layer fully connected neural network with 2048/1024/512 ReLU activated neurons with 0.2 Dropout layer, and an 2-neuron Softmax activation output layer. The training uses the Adam optimizer with categorical crossentropy loss. The model has +2.8M parameters. This classifier represents a generic neural network structure that can be plausibly used by anyone building a neural network-based detector. A more detailed description of the used classifier can be found in [12].

### IV. DATASET

#### A. Data source, ground truth, and flow features

In order to evaluate the ability to detect C2 traffic from the combination of packet sampling and machine learning, this work uses data publicly available from the MTA<sup>1</sup> project, which contains recent, vast, and detailed content related to common malware families. In such a project, various malware infections are described and deployed to run in a sandboxed environment to provide insights on network-based detection. For specific malwares, .pcap files containing captured network traffic associated with the infection and C2 communication are available. After exploring the project's website, it was obtained 508 files corresponding to 20747 TLS, including C2 and non-C2 traffic.

The detection of C2 traffic within this collection resorted to the SSL Blacklist project list of SSL/TLS certificates

<sup>1</sup>MTA - Malware Traffic Analysis: <https://malware-traffic-analysis.net>

employed by botnet C2 servers. It was done by configuring the Suricata IDS with rules to identify malicious TLS flows, which has identified 7672 malicious TLS flows distributed per black-listed malware family certificates including TrickBot, PandaZeus, Gozi, IcedID, and Dridex. The remaining 13075 TLS flows are considered non-malicious traffic, but only part of them was used to create a 50%-50% balanced dataset. This scenario assumes the sampling process runs in a measurement point able to access aggregated traffic, such as a network border router.

Then, Tstat [13] was used to extract 86 numerical features from traffic flows. Incomplete flows were ignored, and non-sampled and sampled flows were normalized independently of each other. Additional details on flow features and normalization can be found in [12].

### B. Sampled vs. non-sampled traffic characteristics

This section provides an overview of the dataset's general characteristics before and after the sampling process. At this point, there is no C2 detection, which is discussed in Section V.

Figure 1 uses a 2D visualization of malicious and normal TLS flows to illustrate the difference between non-sampled and sampled data sets. This visualization was generated by an autoencoder with 2-neuron code, 512-neuron hidden layer, 86 inputs, and 86-neuron output layer, trained with the normalized flow features in the non-sampled training data. Interestingly, the autoencoder seems to push the sampled flows to a region of the 2D space that has almost no non-sampled flows. The autoencoder also seems not to have collapsed all of the sampled flows into a single point of the 2D space, although it is only trained with non-sampled flows.

As expected, sampling does not affect all features alike. Some features are not affected by sampling for the majority of flows. These include features related to counting TCP reset, SYN, FIN, and ACK flags, and to retransmitted and out-of-order packets, as well as timings at the beginning of the stream – such as first TCP payloads and first ACK. Figure 2 identifies the Tstat features whose values changed due to sampling on at least 5% of the flows. These include packet counts, byte counts, total duration and timings of last payloads, and statistics on round trip times, on TCP maximum segment size, and on TCP congestion window.

Figure 2 also shows the boxplot of how much these values changed with sampling. Packet and byte counts boxplots show tens of packets and thousands of bytes being removed from the sampled flows, which seems like a significant impact. In contrast, duration difference median is on the scale of one-tenth of the second. This is a much smaller impact of sampling, and is explainable given that the proposed sampling scheme keeps TCP SYN and FIN/RST packets that allow Tstat to better define the beginning and end of each TCP flow.

Finally, a note about the approach of selecting the first two packets in each direction after TCP handshake. After this process, Tstat is no longer able to detect 7% of the TLS flows including 2.7% of malicious TLS flows. This may be due to

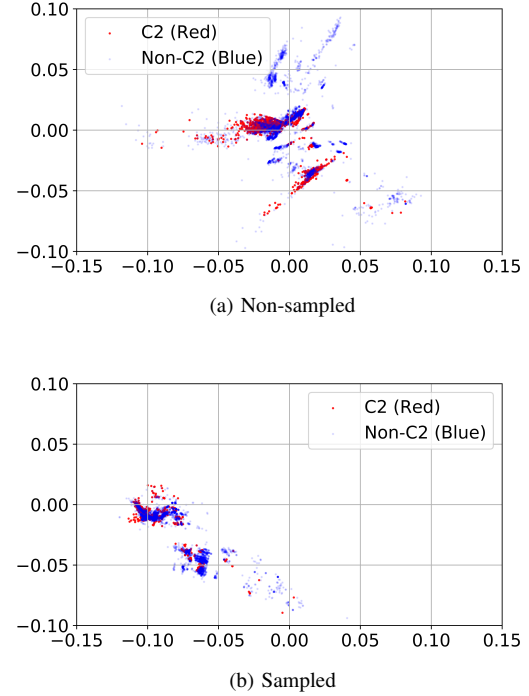


Fig. 1: 2D visualization of non-sampled (a) and sampled (b) data set. Same coordinates and transformation to 2D.

different order in the typical sequence of TLS handshakes, which will be investigated in future work.

## V. RESULTS

The ongoing results are promising in both C2 detection and data reduction. As shown by the boxplots in Figure 3.a, the general accuracy, true positives (malicious flows detected), and true negatives (benign flows detected) of sampled-based detection are only slightly lower than those from non-sampled flows (note the y-scale does not start at 0).

By decreasing the sampling rate from 'Det.-1/3' towards 'Det.-0' (Stage 1-only) there appears to be no consistent degradation of classification performance when compared with analyzing all traffic ('Full'). The benefits of sampling are, nonetheless, significant and increase consistently with sampling rate as expected.

In addition, Figure 3.b shows boxplots for per-flow packet and byte count with different sampling rates. Aggregated, total packets for Stage 1-only sampling is 11.7 times smaller than the number of total packets in the non-sampled dataset (48.1 times smaller for byte counts). Different sampling frequencies in Stage 2 increases packet count in a predictable proportion, due to its deterministic selection pattern.

Considering the accuracy in C2 detection follows the increase in Stage 2 sampling rate, deciding a monitoring strategy should take into consideration the criticality of the monitored link and the available resources for capturing and processing data.

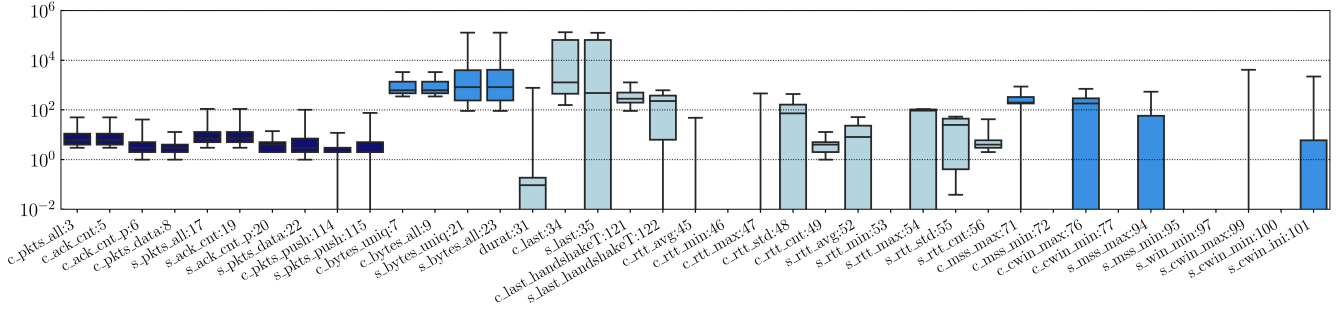
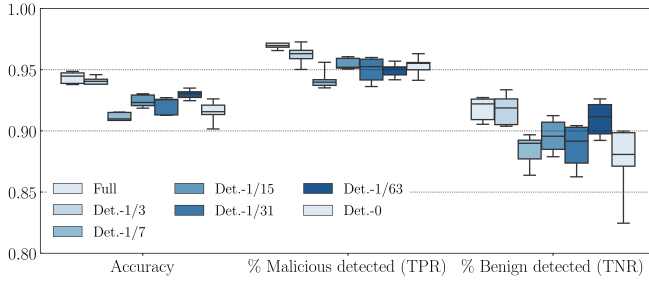
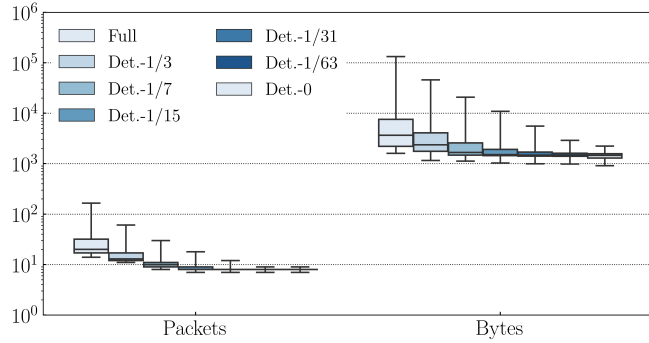


Fig. 2: Boxplot of feature value differences between non-sampled and sampled flows.



(a) Classification results



(b) Effect of sampling

Fig. 3: Ongoing results for multiple sampling rates.

#### A. Detection with original models

Intuitively, the Stage 1-only sampling produces very similar values for the feature *flow duration* because the TCP flow's initial and final packets are always selected for the sample. Prior work [12] has shown that flow duration plays an important role in classifying flows. However, after applying the sampled flows to the model trained with the non-sampled flows, it was observed that classification performance quickly drops to 60% accuracy or below with decreasing sampling rate. It is possible that although duration does not change significantly, the values of the other features cause the original classifier to underperform, whereas the classifiers trained with sampled data are able to take in the new sampled values of the other

features.

This is an interesting finding, as it raises some practical questions about the model to be adopted for training and detection tasks in production networks. For instance, it should be subject of future investigation whether training de machine learning model with a probabilistic technique will degrade the performance of detection based on traffic sampled deterministically, vice versa. It is important to highlight that deciding on a sampling approach is typically done by dynamically balancing resources available, measurement requirements, and network status.

### VI. DISCUSSION ON ALTERNATIVE APPROACHES

#### A. Predicting non-sampled feature values

An alternative to the approach in the previous subsections is to predict feature values of non-sampled traffic from the feature values of sampled traffic, and then apply the detector that has been trained with non-sampled traffic. To build a non-sampled feature predictor, a dense neural network was trained with 86 inputs, a 2048 neuron densely connected, *tanh* activated intermediate layer, and an 86 neurons, densely connected, LeakyReLU activated output layer. The size of the input and output corresponds to the dimension of feature values – sampled feature values on the input and non-sampled feature values of the same flow on the output. The training uses the Adam optimizer with *logcosh* loss.

In Figure 4, the same 2D visualization is applied as before to the non-sampled predictions. It can be observed that although the visualization is different from the visualization of non-sampled data in Figure 1.a, the predicted points are now centered, much as the non-sampled points are and unlike sampled points in Figure 1.b. This indicates the predictions are closer to the non-sampled features than the sampled features. However, it can also be observed that the non-C2 traffic is less dispersed than the original non-sampled, non-C2 traffic. This may be an indicator of the degraded classification performance obtained by feeding predicted values to the C2 detector.

In fact, classification results are worse than with the approach where feature values from sampled traffic are used for training: it is obtained a low score of 65% accuracy, with only 33% true negatives yet 97% true positives. Future work aims at improving these results by including different settings for

the neural network that predicts non-sampled from sampled feature values.

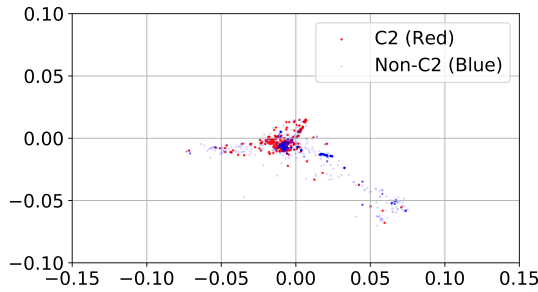


Fig. 4: Non-sampled traffic predicted from sampled traffic.

### B. Learning which packets to sample

As discussed in Section II, different sampling techniques can be configured to meet specific measurement requirements and resource constraints. Typically, this setup is done beforehand by administrators with few or no flexibility. More recently, adaptive techniques have introduced the ability to change sampling patterns in runtime [2].

Although being a clear step up compared to the classical approach, adaptive techniques still rely on some kind of parameterization introduced by the network administrators. For instance, the Multiadaptive technique [14] requires previous configurations regarding the minimum and maximum sample size and algorithm reactivity.

Such a scenario opens opportunities for a third alternative approach, in which machine learning provides dynamic inputs for both classical and adaptive sampling techniques. This may be useful in any monitoring task assisted by packet sampling, though it is particularly sensitive for security-related activities due to the highly dynamic nature of different malicious traffic.

In this way, future work aims at investigating the effectiveness of this approach and its compared performance with those presented along this work.

## VII. CONCLUSION

Sampling is a necessity in some current networks and comes with a price tag of underperformance unless carefully designed for specific network management tasks. This paper looked at sampling for the task of detection of encrypted command and control traffic. The first approach at sampling packets irrespectively of flows resulted in no TLS flow being detected by Tstat and thus it was chosen to sample packets deterministically in each flow. Then, a dense neural network classifier was trained with sampled feature values and were able to detect C2 traffic without much classification performance loss, for a varying range of sampling rate.

It is important to highlight that such high accuracy in C2 traffic detection was achieved with a significant reduction in data analysed. Although being early results, it is clear that packet sampling can be successfully used for security-related

tasks, which has been a common subject of interest by the scientific community.

Future work aims to improve these results by exploring new sampling approaches that can sustain the classifier performance at non-sampled levels and improve the prediction of non-sampled feature values from sampled feature values. This may be a valuable strategy toward supporting different network management tasks without requiring a specific sampling approach for each task. It also intended to work on the dataset's limitations, including adding more normal traffic and making it more unbalanced, like what would happen in a real network. Finally, another interesting approach will consist of exploring to which extent machine learning can be used to adjust sampling dynamically, i.e., through adaptive techniques.

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### REFERENCES

- [1] B. Claise, A. Johnson, and J. Quittek, "Packet Sampling (PSAMP) Protocol Specifications RFC 5476," IETF, Tech. Rep., 2009.
- [2] J. M. C. Silva, P. Carvalho, and S. R. Lima, "A modular traffic sampling architecture: bringing versatility and efficiency to massive traffic analysis," *Journal of Network and Systems Management*, vol. 25, no. 3, pp. 643–668, 2017.
- [3] N. Duffield *et al.*, "Sampling for passive internet measurement: A review," *Statistical Science*, vol. 19, no. 3, pp. 472–498, 2004.
- [4] A. Salama, R. Saatchi, and D. Burke, "Adaptive sampling for QoS traffic parameters using fuzzy system and regression model," *Mathematical Models and Methods in Applied Sciences*, vol. 11, pp. 212–220, 2017.
- [5] D. Brauckhoff, B. Tellenbach, A. Wagner, M. May, and A. Lakhina, "Impact of packet sampling on anomaly detection metrics," in *Proceedings of the 6th ACM SIGCOMM conference on Internet measurement*, 2006, pp. 159–164.
- [6] N. Z. Bawany, J. A. Shamsi, and K. Salah, "DDoS attack detection and mitigation using SDN: methods, practices, and solutions," *Arabian Journal for Science and Engineering*, vol. 42, no. 2, pp. 425–441, 2017.
- [7] T. Ha, S. Kim, N. An, J. Narantuya, C. Jeong, J. Kim, and H. Lim, "Suspicious traffic sampling for intrusion detection in software-defined networks," *Computer Networks*, vol. 109, pp. 172–182, 2016.
- [8] B. Anderson and D. McGrew, "Machine Learning for Encrypted Malware Traffic Classification: Accounting for Noisy Labels and Non-Stationarity," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ser. KDD '17. New York, NY, USA: ACM, 2017, pp. 1723–1732.
- [9] G. Marín, P. Casas, and G. Capdehourat, "Deep in the Dark - Deep Learning-Based Malware Traffic Detection Without Expert Knowledge," in *2019 IEEE Security and Privacy Workshops (SPW)*, May 2019.
- [10] M. Mellia, A. Carpani, and R. L. Cigno, "Tstat: Tcp statistic and analysis tool," in *International Workshop on Quality of Service in Multiservice IP Networks*. Springer, 2003, pp. 145–157.
- [11] B.-Y. Choi, J. Park, and Z.-L. Zhang, "Adaptive packet sampling for accurate and scalable flow measurement," in *IEEE Global Telecommunications Conference, 2004. GLOBECOM '04.*, vol. 3, Nov. 2004, pp. 1448–1452 Vol.3.
- [12] C. Novo and R. Morla, "Flow-based detection and proxy-based evasion of encrypted malware c2 traffic," in *Proceedings of the 13th ACM Workshop on Artificial Intelligence and Security*, ser. AISec'20. New York, NY, USA: Association for Computing Machinery, 2020, p. 83–91. [Online]. Available: <https://doi.org/10.1145/3411508.3421379>
- [13] M. Trevisan, A. Finamore, M. Mellia, M. Munafo, and D. Rossi, "Traffic Analysis with Off-the-Shelf Hardware: Challenges and Lessons Learned," *IEEE Commun. Mag.*, vol. 55, no. 3, pp. 163–169, 2017.
- [14] J. M. C. Silva, P. Carvalho, and S. R. Lima, "A multiadaptive sampling technique for cost-effective network measurements," *Computer Networks*, vol. 57, no. 13, pp. 3357–3369, 2013.